

MD KASHIF ALAM

Robotics Software Engineer

CONTACT & PORTFOLIO

Email: mdkashifalam376@gmail.com

+91 9142780681

Portfolio: kashifalam407.github.io/Portfolio/

LinkedIn: linkedin.com/in/md-kashif-alam-a55b19380

GitHub: github.com/KashifAlam407

YouTube: www.youtube.com/@electroboticsAi

TECHNICAL SUMMARY

I am a robotics-focused student associated with IIT Patna, with hands-on experience in autonomous systems and robotics software development. My background includes sustained project-based learning, progressing from custom electronics to system-level robotics. I document technical experiments and implementations through my **portfolio** and my Youtube **Electrobotics AI** platform. My work centers on integrating embedded firmware with ROS2-based robotic systems, emphasizing robust hardware–software integration and real-world system behavior.

EDUCATION

- **Bachelor of Science (BS)**, Artificial Intelligence and CyberSecurity | IIT Patna

TECHNICAL SKILLS

- **Robotics Frameworks:** ROS2 (Humble), Nav2 Stack, MoveIt2, URDF Modeling, TF2, SLAM
- **Programming Languages:** C++ (OOP), Python, MicroPython, Bash Scripting
- **Computer Vision & AI:** OpenCV, YOLO, MediaPipe, TensorFlow
- **Embedded Systems:** ESP32, STM32, Raspberry Pi, Arduino, PCA9685
- **Control & Kinematics:** PID Tuning, Inverse Kinematics, Gait Dynamics, State Estimation
- **Hardware Communication:** UART (Serial), I2C, SPI, CAN Bus (Basics)
- **Simulation & Modeling:** Gazebo, RViz2, Fusion 360 (CAD), MATLAB/Simulink, OpenVSP
- **Development Tools:** Linux (Ubuntu), Git/GitHub, VS Code, WSL2, Docker

INDUSTRY EXPERIENCE

- **Technical Intern — India Space Lab (ISL)** Jan 2026 – Present | Remote
- **AI & LLM Engineer Intern – Springer Capital** Feb 2026 – Present | Remote

TECHNICAL PROJECTS

- **Industrial Robot Setup – Fanuc CRX-10iA (ROS2 & MoveIt)**
 - Configured Fanuc CRX-10iA with MoveIt 2 for motion planning and trajectory execution.
 - *GitHub:* <https://github.com/KashifAlam407/fanuc-crx10ia-moveit-task1>
- **Custom DWA Local Planner (ROS2)**
 - Developed a custom DWA-based local planner in ROS 2 for obstacle-aware motion planning using LiDAR and odometry.
 - *GitHub:* https://github.com/KashifAlam407/Custom_DWA_Planner-ROS2
- **Autonomous Mobile Robot (AMR)**
 - Built an AMR using Raspberry Pi and RPLIDAR with ROS 2, capable of basic path planning and autonomous navigation.
 - *GitHub:* <https://github.com/KashifAlam407/AMR>
- **Quadruped Robotic Dog**
 - Engineering an autonomous quadruped robot capable of walking, running, and turning, with future integration of AI-based navigation, mapping, and localization.
 - *GitHub:* https://github.com/KashifAlam407/Quadruped_Robot
- **AI-Powered Gesture-Controlled Robotic Hand**
 - Developed a robotic hand controlled via OpenCV and Mediapipe.
 - *GitHub:* https://github.com/KashifAlam407/Gesture_Controlled
- **Surveillance Drone**
 - Designed and built a drone equipped with real-time video streaming for aerial monitoring and surveillance.
 - *GitHub:* https://github.com/KashifAlam407/Drone_Quadcopter